



Hierarchical Star Formation in the 30 Doradus Region, the Orion Nebula Cluster and Westerlund 1

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Abstract

- 30 Dor region is a great target for hierarchical star formation studies because the object is an actively star-forming region (Sun et al., 2017).
- Hierarchical star formation is star formation without a preferred spatial scale.
- It appears to be driven by supersonic turbulence in the interstellar medium (ISM)
- Binary star clusters seem to be overrepresented in galaxies; is that a real effect?

Main aim: To investigate the hierarchical star formation of 30 Doradus (30 Dor) in the Large Magellanic Cloud (LMC). This includes identifying stellar and cluster ages and spatial distributions.

- The fractal dimension (FD) and virial ratio (α_{vir}) are key. Sun et al. (2017) selected Upper Main Sequence (UMS) stars from the color-magnitude diagram (CMD).
- To compare UMS and old stars, a new CMD was produced, including the UMS selection from Sun et al. (2017). Red Giant Branch (RGB) and Red Clump stars were selected as older comparison sample.
- The Orion Nebula Cluster (ONC) and Westerlund 1 (Wd1), projected to the LMC distance, were selected for realistic comparison.

CMD Results

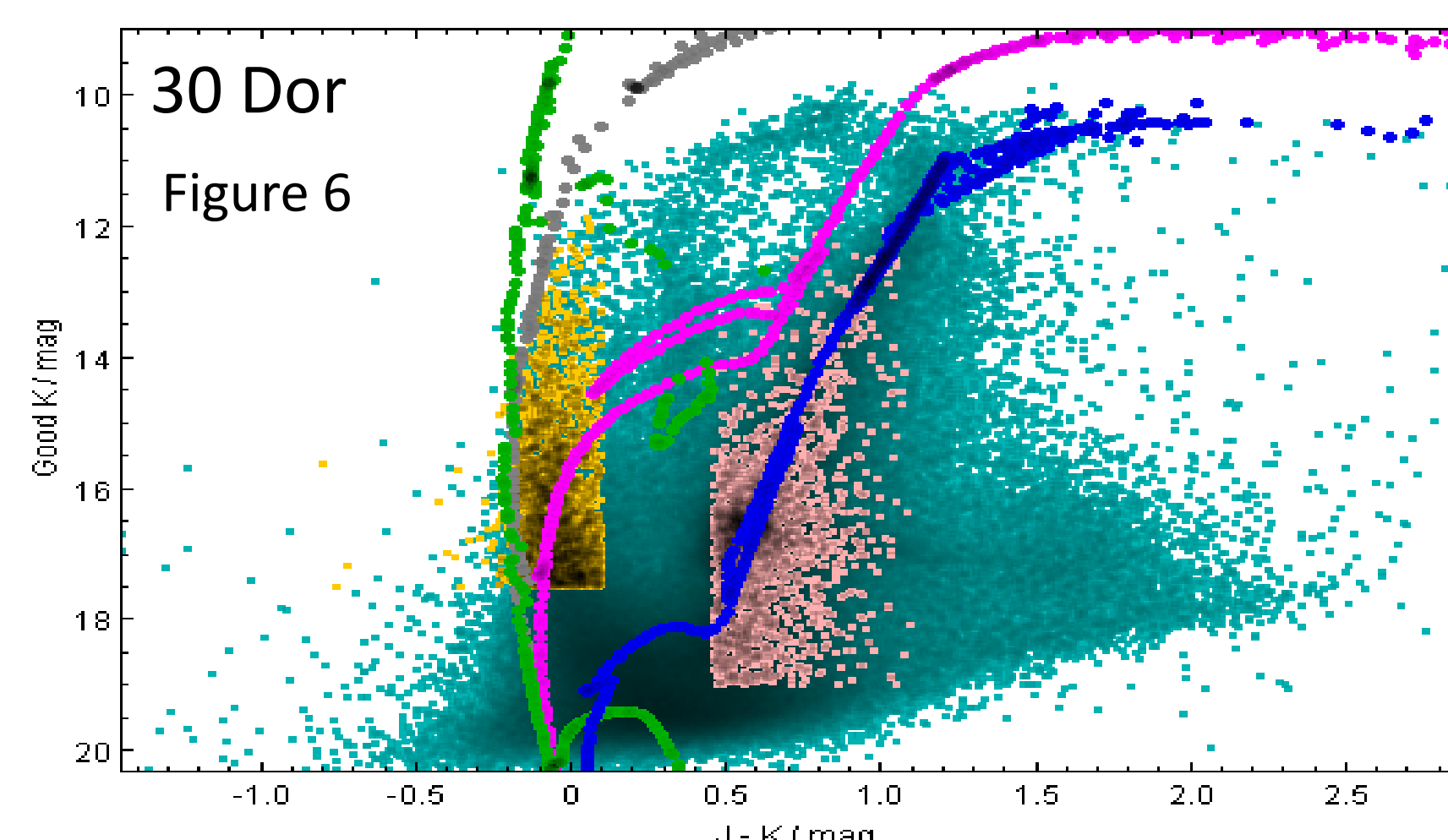
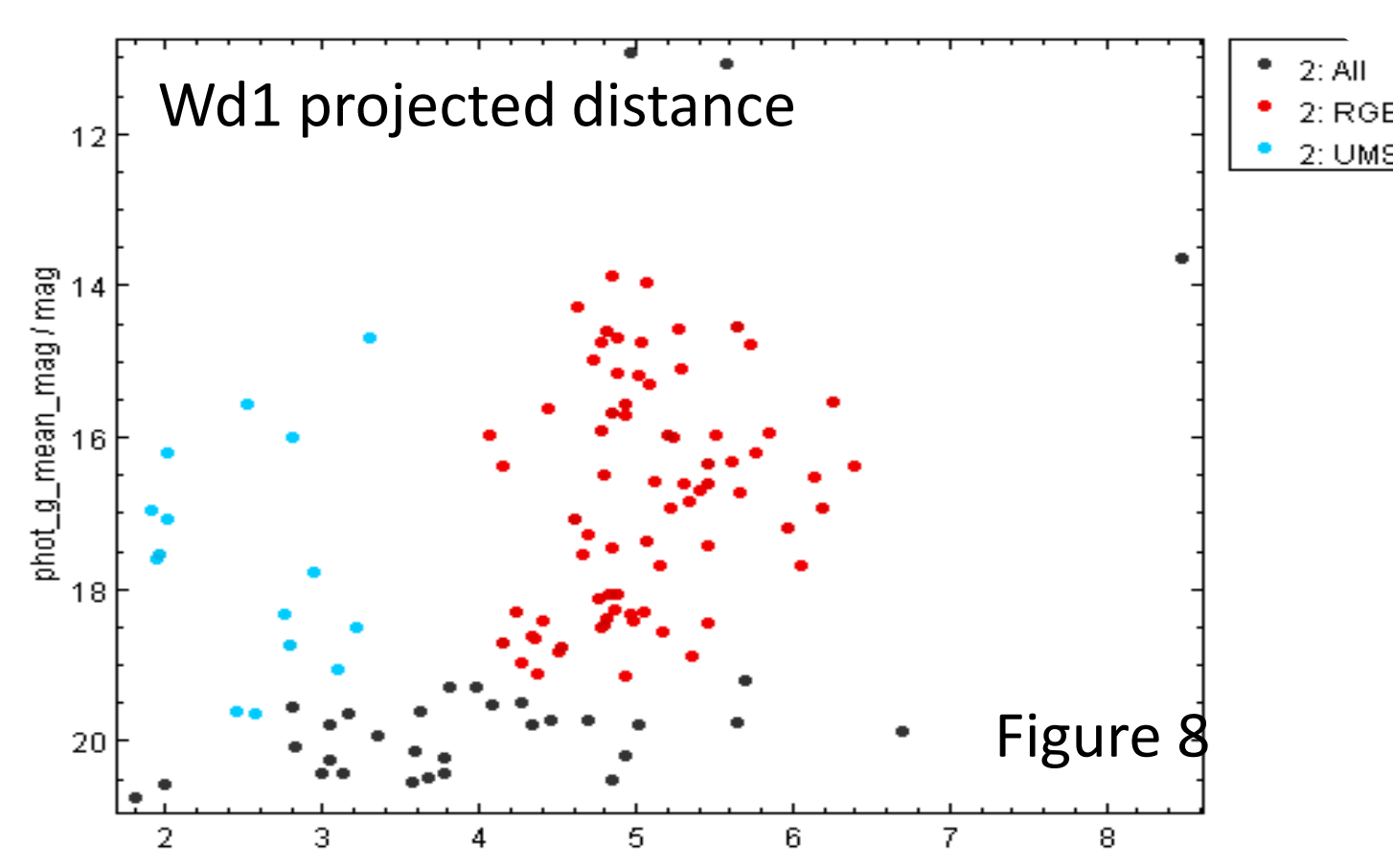
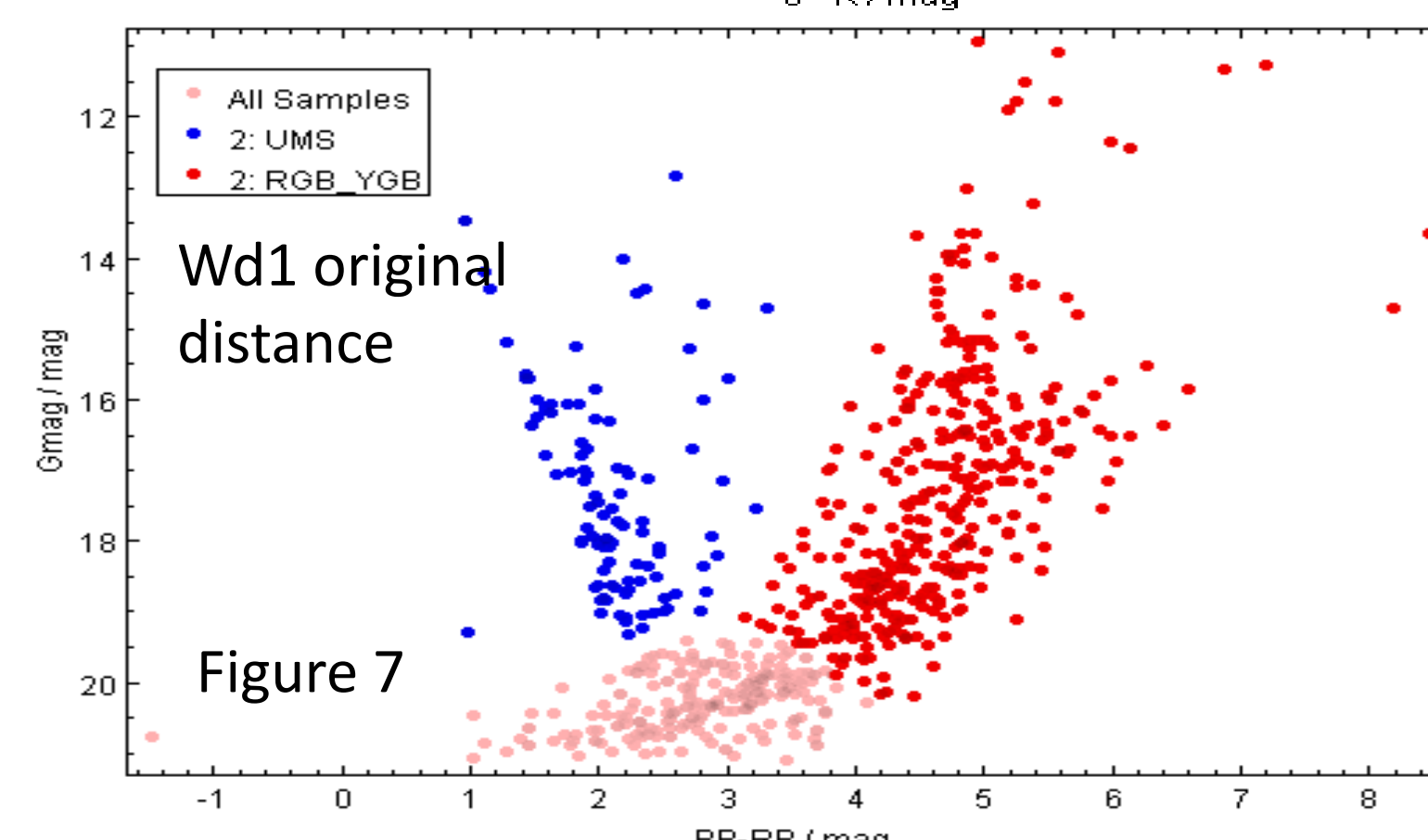


Figure 6 (top): CMD for the 30 Dor region with isochrones $\log(\tau/\text{yr}) = 6.6, 7, 8 \text{ \& } 9$

Figure 7 (bottom left): CMD for Wd1 at the original distance with UMS & RGB stars.

Figure 7 (bottom right): CMD for Wd1 at the LMC's distance with UMS & RGB stars.



Fractal Dimension Results

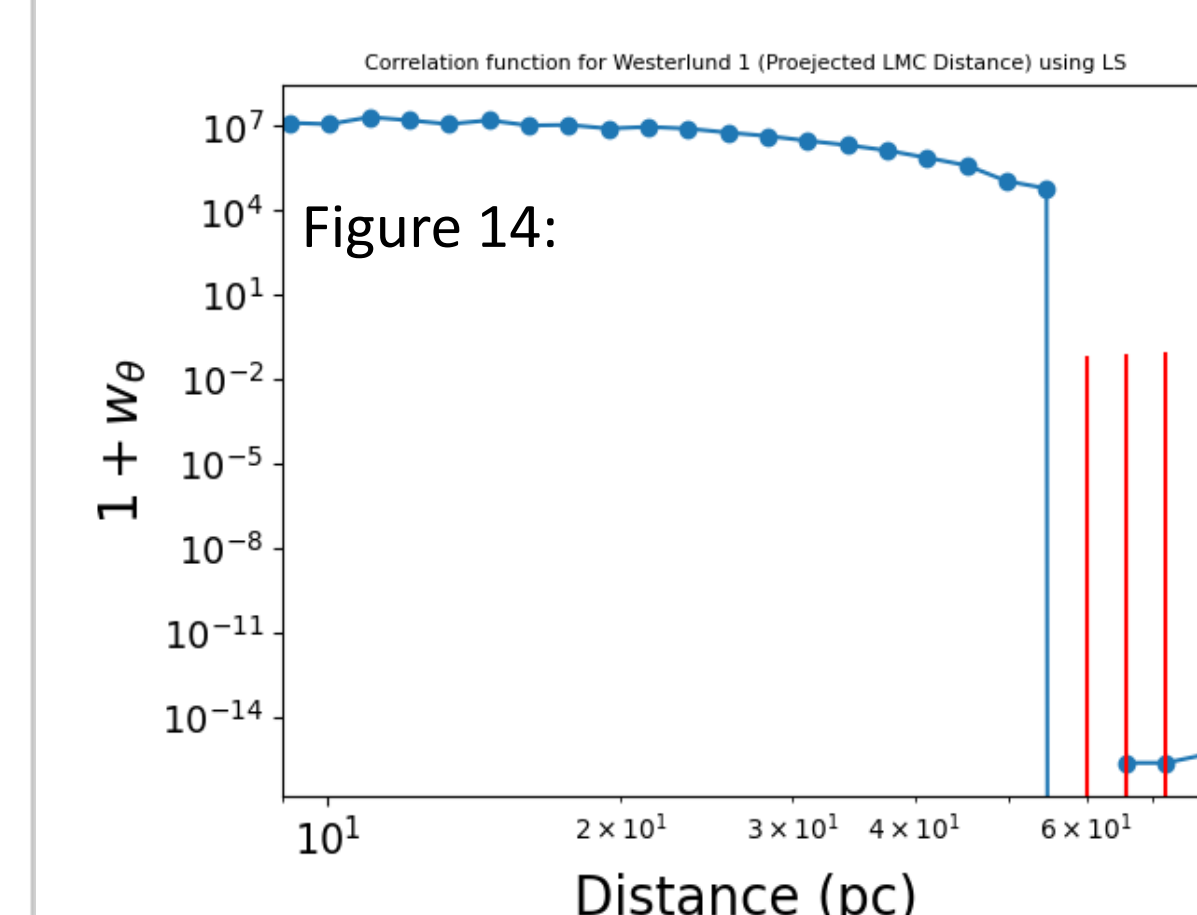
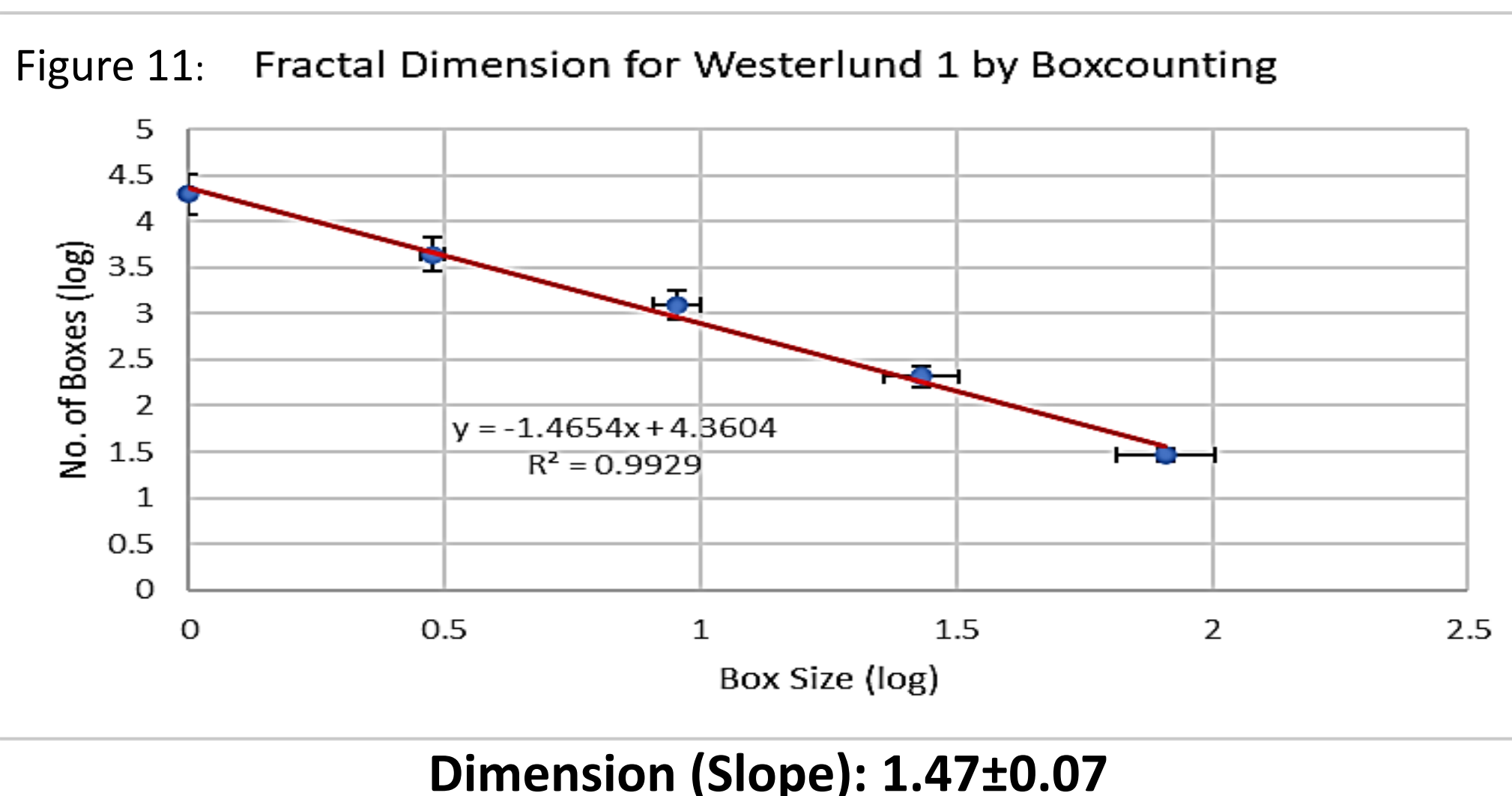
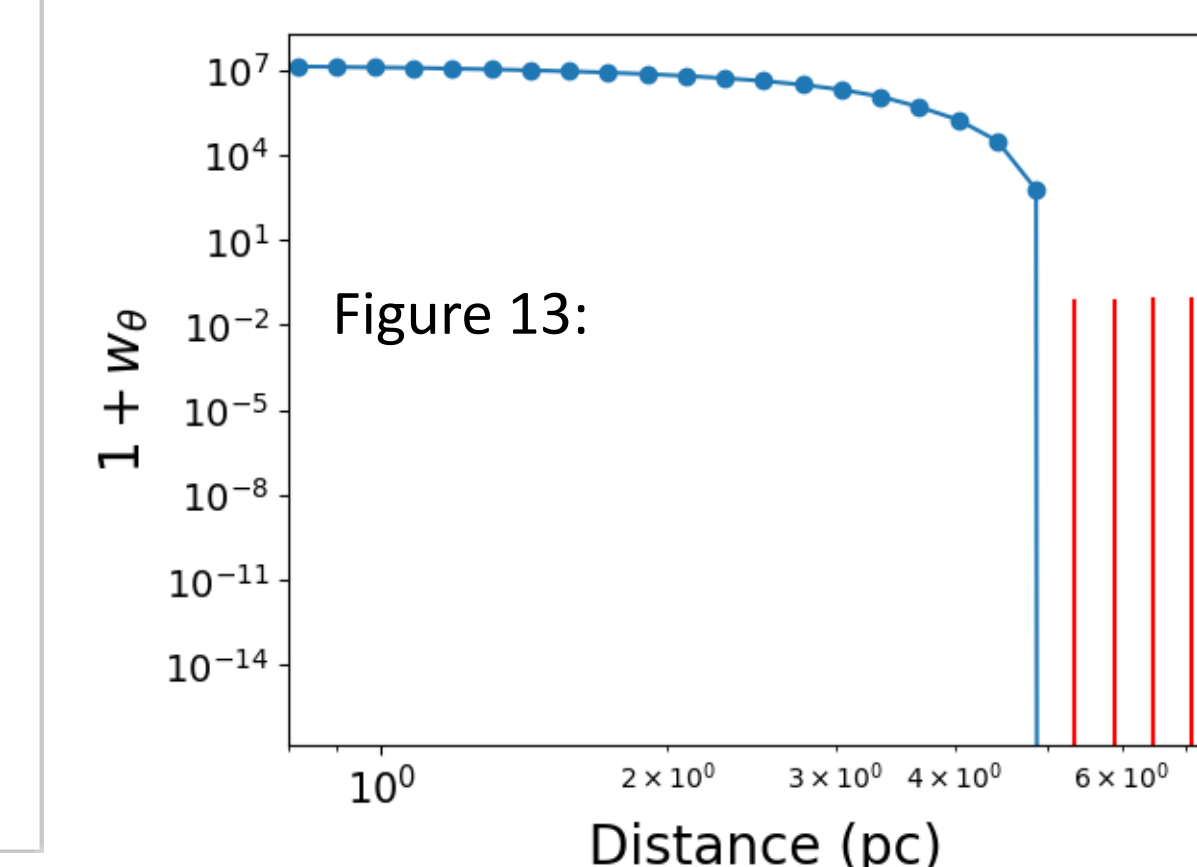
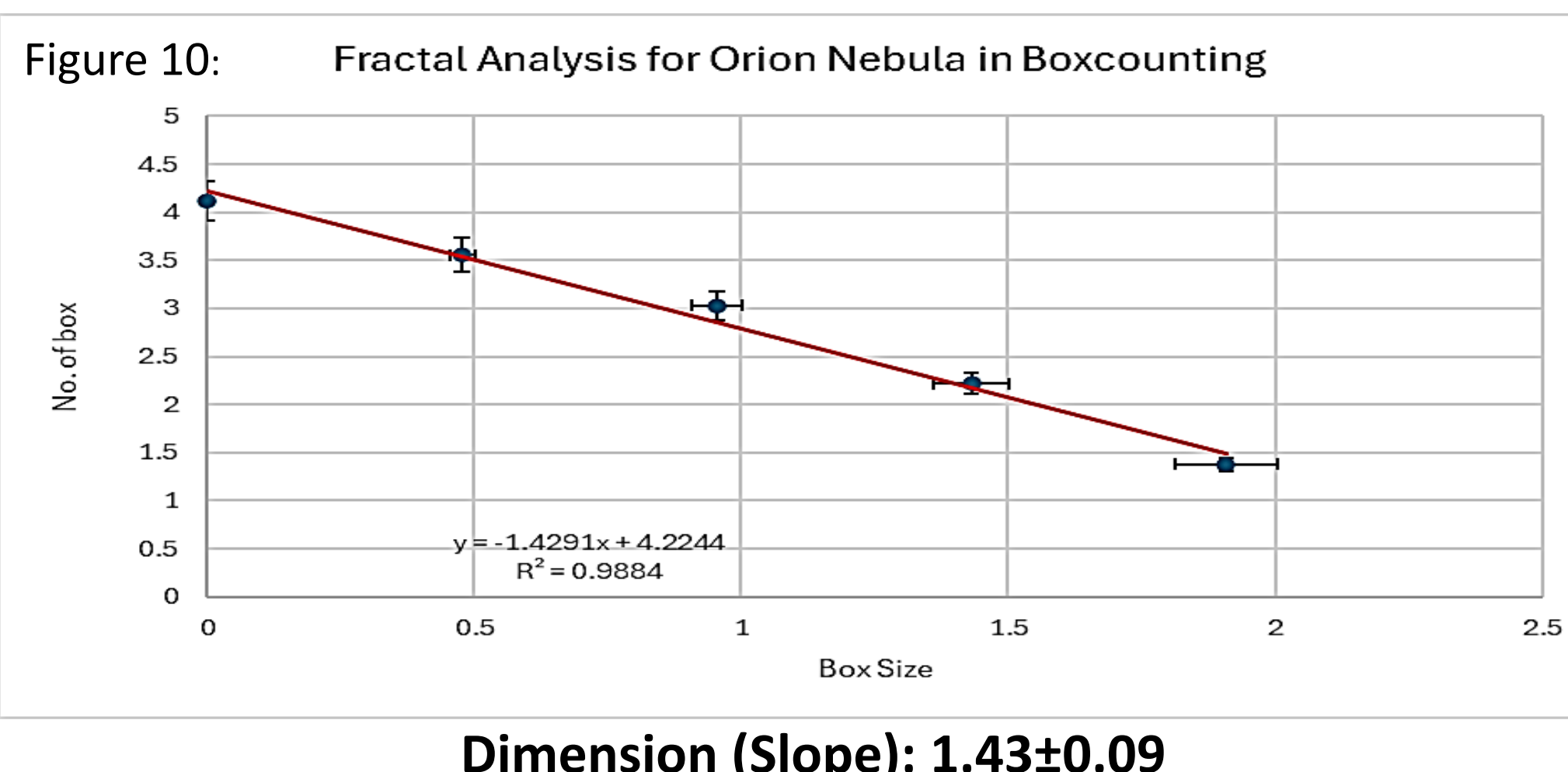
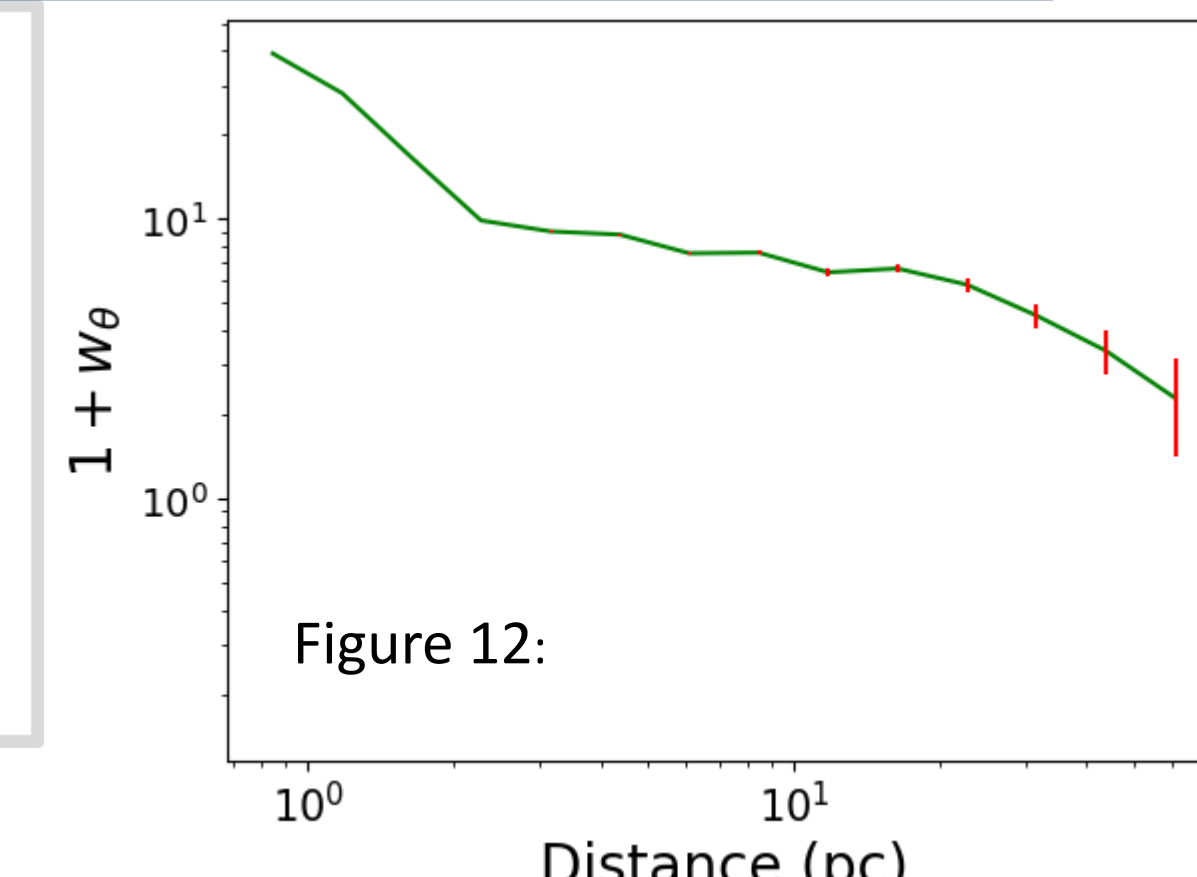
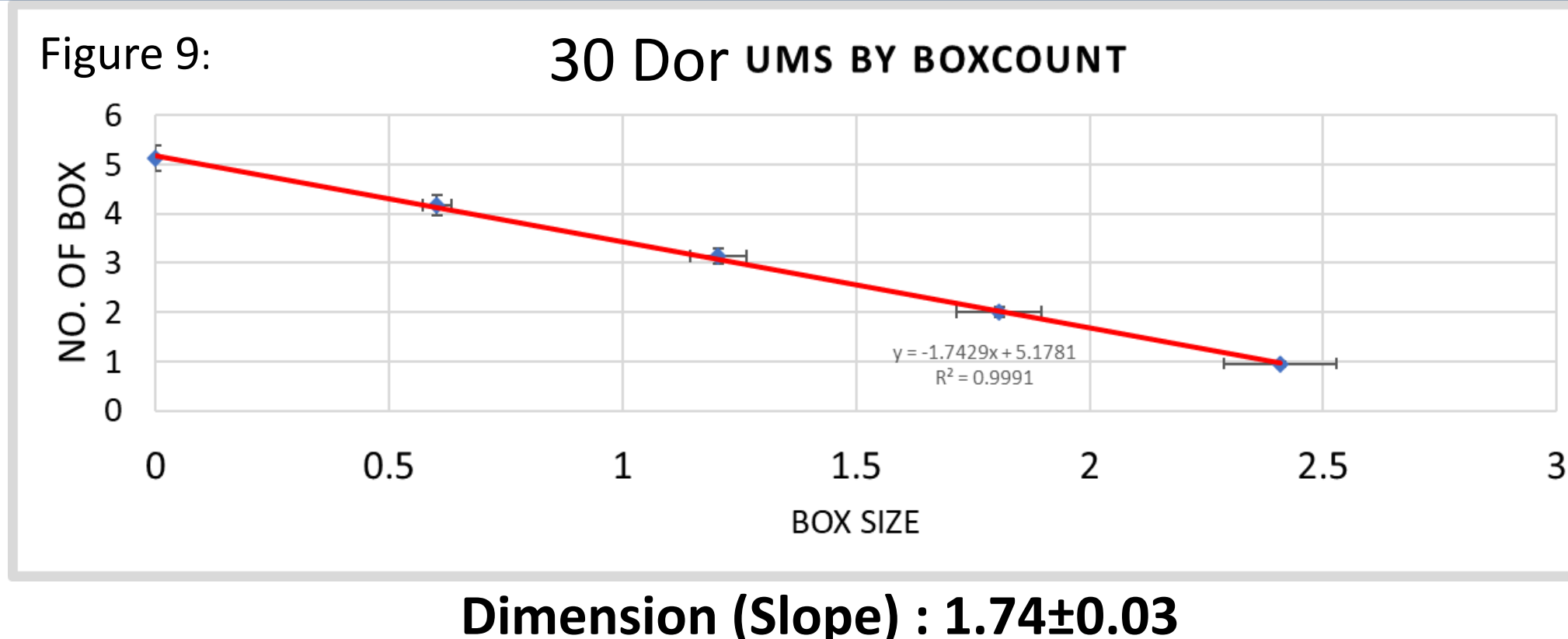


Figure 9 (top): Fractal dimension analysis for 30 Dor
Figure 10 (middle): Fractal dimension analysis for the ONC
Figure 11 (bottom): Fractal dimension analysis for Wd1 (Original distance)

2-point correlation for 30 Dor (top, figure 12), the Wd1 Original Distance (middle, figure 12), and Wd1 projected LMC Distance (bottom, figure 13).

Conclusions

- 30 Dor, the ONC and Wd1 have very similar values of FD at their original distances, ranging from 1.43 to 1.74 (highly substructured and clumpy for clusters).
- The ONC projected at the LMC distance has a very low resolution and few stars detected; it would be unidentifiable as a star cluster.
- The Minimum Spanning Tree (MST) helps achieve better results for the FD.
- The 2-point correlation function (2PCF) achieves similar results for 30 Dor, ONC and Wd1 (*work in progress*).
- Wd1 can still be identified as a star cluster projected at the LMC distance.

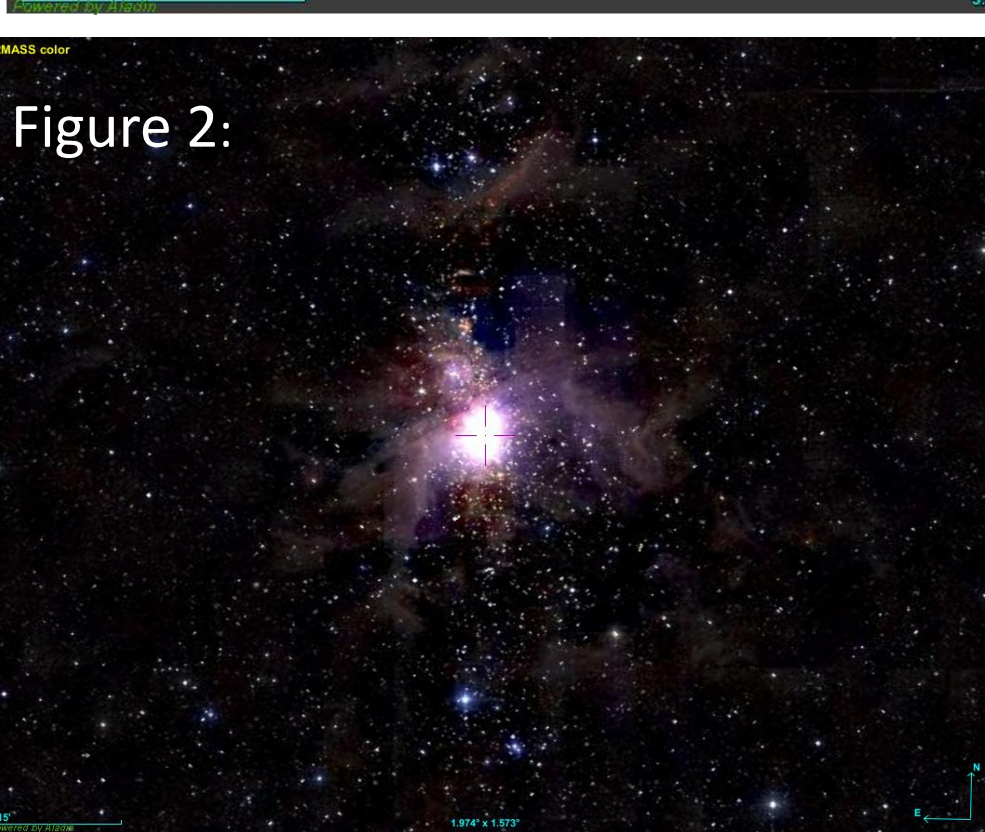
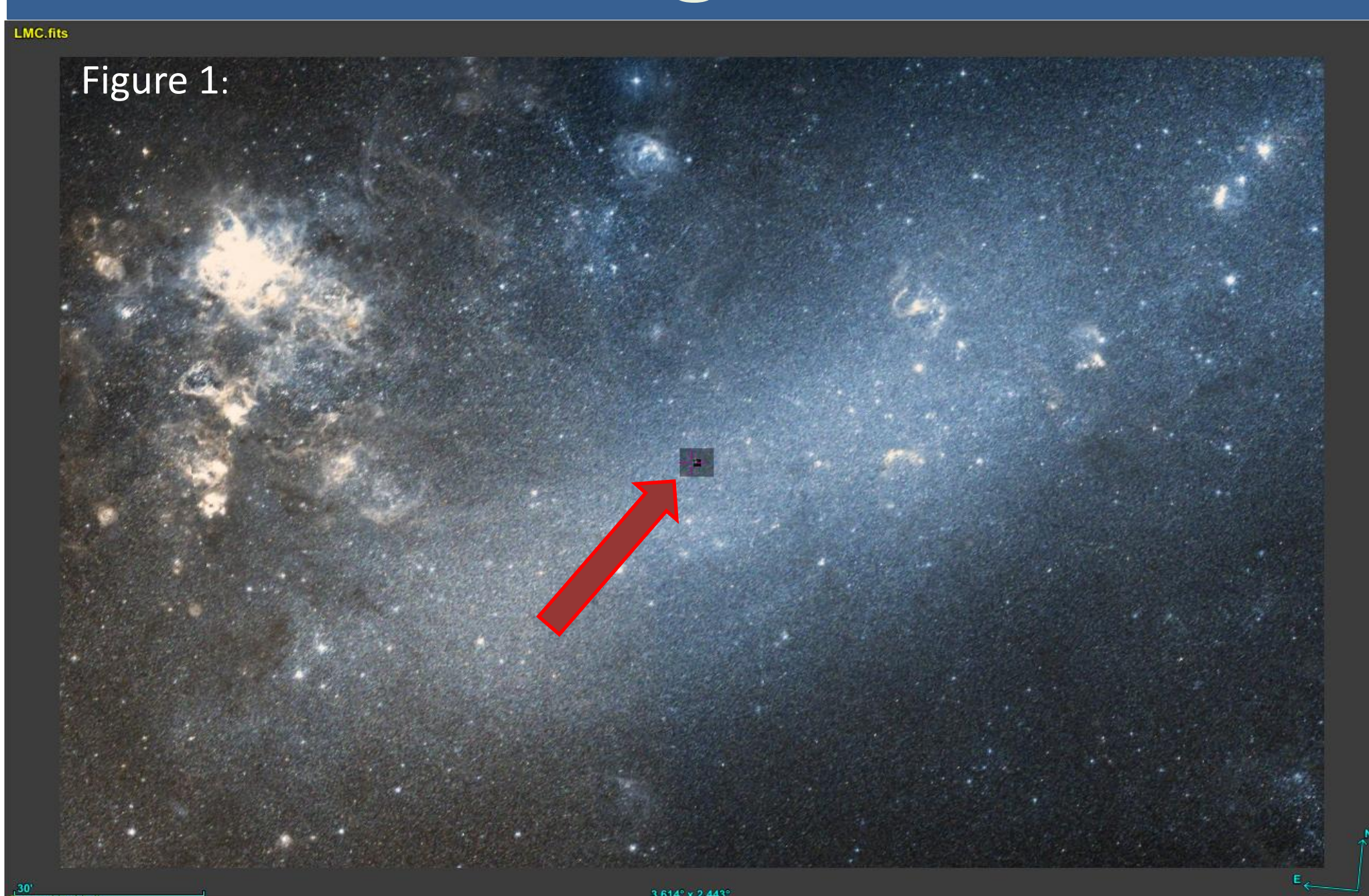
Background

- The FD measures the number of points in a sample; it is represented by a non-integer number. N is the number of samples while r is a scaling factor:

$$D = \frac{\log(N)}{\log(r)}$$

- The virial ratio measures the equilibrium between kinetic energy (T) and gravitational potential energy (Ω): $\alpha_{\text{vir}} = T/|\Omega|$

Targets



Detection

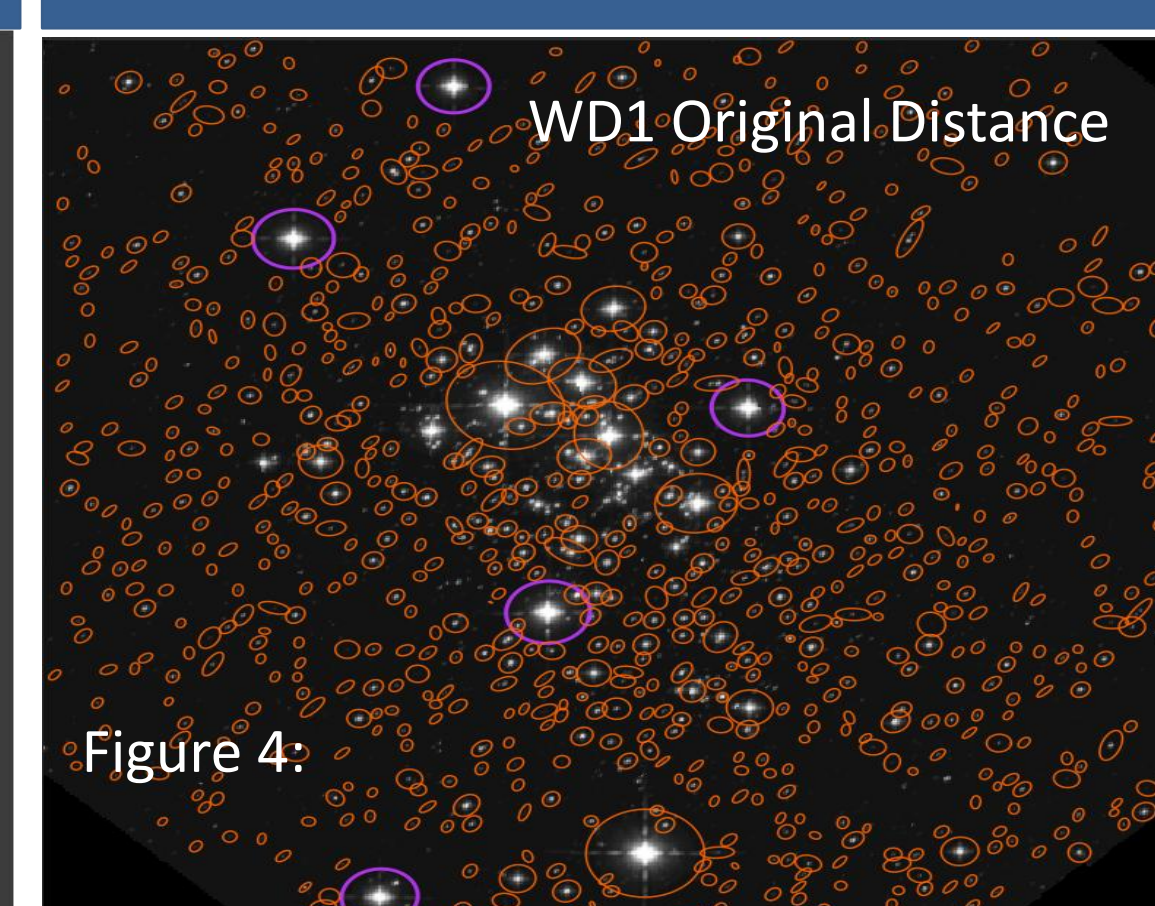


Figure 1 (top). 30 Doradus (top left), the most active LMC star-forming region; Wd1 and the ONC, projected to the LMC distance are shown by a red arrow (DSS2 color). **Figure 2 (bottom left):** Orion Nebula (2MASS IR); **Figure 3 (bottom right):** Westerlund 1 (Hubble Space Telescope, F160W)

Figures 4 (top): Detection of Wd1 stars at original distances.

Figure 5 (bottom): Detection of Wd1 stars at projected (LMC) distances.

Note: Detections of ONC stars at the LMC distance are too few to form a star cluster.

Physical Explanation of the 2PCF & MST

- 2PCF works by measuring the probability of two stars being located at a given distance.
- This function helps trace the amplitude of stellar clustering.
- MST connects stars within a cluster with a minimum distance.
- It also represents the basis of the cluster's connectivity, revealing the hierarchical organization and relationships between the stars.

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